

# MA4110/MA5020: Computational Methods for Fluid Flow

## Tutorial Sheet 2

Finite Difference Approximations, Stability Analysis, and Numerical Errors

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### Question 1: Order of Accuracy of Finite Difference Formulas

- (a) Using Taylor series expansions about  $x_j$ , find the leading truncation error and order of accuracy of the one-sided backward difference formula:

$$u'(x_j) \approx \frac{3u_j - 4u_{j-1} + u_{j-2}}{2\Delta x}$$

- (b) Use the method of undetermined coefficients to find constants  $a$  and  $b$  such that the central stencil

$$u'(x_j) \approx \frac{a(u_{j+1} - u_{j-1}) + b(u_{j+2} - u_{j-2})}{\Delta x}$$

is fourth-order accurate, i.e., has truncation error  $\mathcal{O}(\Delta x^4)$ .

### Question 2: Modified Equation Analysis: Why FTCS Blows Up

Consider the explicit FTCS scheme applied to the linear advection equation  $u_t + au_x = 0$  ( $a > 0$ ):

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

- (a) Expand  $u_j^{n+1}$  and  $u_{j\pm 1}^n$  in Taylor series about  $(x_j, t^n)$  up to second order in  $\Delta t$  and  $\Delta x$ .  
(b) Eliminate the time derivative  $u_{tt}$  using the governing PDE  $u_t = -au_x$  ( $u_{tt} = a^2 u_{xx}$ ) to derive the modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = -\frac{a^2 \Delta t}{2} \frac{\partial^2 u}{\partial x^2}$$

- (c) From physical principles of diffusion, explain why a **negative diffusion coefficient** causes catastrophic instability for any  $\Delta t > 0$ .

### Question 3: Hand Calculation: 1D Advection with a Step Profile

Consider the advection equation  $\frac{\partial u}{\partial t} + 2 \frac{\partial u}{\partial x} = 0$  on  $x \in [0, 1]$  with  $\Delta x = 0.2$  (nodes  $x_0 = 0.0, x_1 = 0.2, \dots, x_5 = 1.0$ ) and  $\Delta t = 0.05$ .

Initial condition:  $u_0^0 = u_1^0 = u_2^0 = 1.0$  and  $u_3^0 = u_4^0 = u_5^0 = 0.0$ , with boundary condition  $u(0, t) = 1.0$ .

- (a) Calculate the Courant number  $\nu = \frac{a\Delta t}{\Delta x}$ . Is the First-Order Upwind scheme stable?  
(b) Using the upwind formula  $u_j^{n+1} = (1 - \nu)u_j^n + \nu u_{j-1}^n$ , compute the numerical solution at:
  - $t = 0.05$  ( $n = 1$ ) for all nodes  $j = 0, 1, 2, 3, 4, 5$ .
  - $t = 0.10$  ( $n = 2$ ) for all nodes  $j = 0, 1, 2, 3, 4, 5$ .(c) Compare your computed values at  $x_2 = 0.4$  and  $x_3 = 0.6$  with the exact sharp step to observe how numerical diffusion smears the shock.

**Question 4: Hand Calculation: 1D Diffusion Equation (FTCS)**

Solve  $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$  on  $x \in [0, 1]$  with boundary conditions  $u(0, t) = 0$ ,  $u(1, t) = 0$  using explicit FTCS with  $\Delta x = 0.2$  and  $\Delta t = 0.02$ .

Initial values at  $x = [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]$  are:  $u^0 = [0, 10, 20, 20, 10, 0]$ .

- Calculate the diffusion number  $r = \frac{\Delta t}{\Delta x^2}$ . Is this computation numerically stable?
- Show that when  $r = 0.5$ , the FTCS scheme simplifies to  $u_j^{n+1} = \frac{1}{2}(u_{j+1}^n + u_{j-1}^n)$ .
- Using this formula, compute the solution at the internal nodes ( $j = 1, 2, 3, 4$ ) at:
  - Step 1:  $t = 0.02$
  - Step 2:  $t = 0.04$

**Question 5: Stability Analysis: Backward-Time Backward-Space (BTBS)**

Consider the implicit upwind scheme for the advection equation  $u_t + au_x = 0$  ( $a > 0$ ):

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_j^{n+1} - u_{j-1}^{n+1}}{\Delta x} = 0$$

- Substitute a Fourier mode  $u_j^n = \hat{u}^n e^{ij\xi}$  and derive the amplification factor  $G(\xi)$  in terms of  $\nu = \frac{a\Delta t}{\Delta x}$  and  $\xi = k\Delta x$ .
- Show that  $|G(\xi)|^2 = \frac{1}{(1 + \nu - \nu \cos \xi)^2 + (\nu \sin \xi)^2}$ .
- Prove that  $|G(\xi)| \leq 1$  for all  $\xi \in [-\pi, \pi]$  for **any**  $\nu > 0$ . What does this imply about the stability restriction on  $\Delta t$ ?

**Question 6: Stability Analysis: The Leapfrog (CTCS) Scheme**

The Leapfrog scheme for  $u_t + au_x = 0$  is central in both time and space:

$$\frac{u_j^{n+1} - u_j^{n-1}}{2\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

- Substitute  $u_j^n = G^n e^{ij\xi}$  and show that  $G$  satisfies:  $G^2 + 2i\nu \sin \xi G - 1 = 0$ .
- Solve for the roots  $G_{1,2} = -i\nu \sin \xi \pm \sqrt{1 - \nu^2 \sin^2 \xi}$ .
- Show that when  $|\nu| \leq 1$ , both roots satisfy  $|G| = 1$  identically. What does  $|G| = 1$  imply about numerical dissipation?

**Question 7: Phase Velocity and Dispersion Error Calculation**

A dispersive scheme for  $u_t + au_x = 0$  (physical wave speed  $a = 2.0$  m/s) has the modified equation:

$$\frac{\partial u}{\partial t} + 2 \frac{\partial u}{\partial x} = -C \frac{\partial^3 u}{\partial x^3}, \quad \text{where } C = 0.02 \text{ m}^3/\text{s}$$

The phase velocity for wavenumber  $k$  is given by  $c(k) = a - Ck^2$ .

- (a) Calculate the numerical phase speed  $c(k)$  for:
  - A long-wavelength mode with  $k = 1.0$  rad/m.
  - A short-wavelength mode with  $k = 6.0$  rad/m.
- (b) Which mode travels slower than the physical wave speed?
- (c) Because short-wavelength ripples travel slower than the main front, do spurious oscillations appear **ahead of** (leading) or **behind** (trailing) the wave front?

**Question 8: CFL Condition on a Multi-Zone Grid**

A 1D flow simulation ( $a = 2.5$  m/s) runs across two contiguous mesh zones:

- **Zone 1 (Coarse region):**  $x \in [0, 1.0]$  with grid spacing  $\Delta x_1 = 0.05$  m.
- **Zone 2 (Refined boundary layer):**  $x \in [1.0, 1.5]$  with grid spacing  $\Delta x_2 = 0.01$  m.

If an explicit scheme requires  $\nu \leq 0.8$  everywhere for stability:

- (a) Find the maximum allowable time step  $\Delta t_{\max}$  that guarantees stability across the *entire* computational domain.
- (b) Which zone limits the global time step? What is the effective CFL number in the coarse zone when using this  $\Delta t_{\max}$ ?

**Question 9: Monotonicity Preserving Schemes and Godunov's Barrier**

Consider two explicit linear schemes of the form  $u_j^{n+1} = c_{-1}u_{j-1}^n + c_0u_j^n + c_1u_{j+1}^n$ :

$$\text{Scheme A: } u_j^{n+1} = \frac{1}{4}u_{j-1}^n + \frac{1}{2}u_j^n + \frac{1}{4}u_{j+1}^n, \quad \text{Scheme B: } u_j^{n+1} = -\frac{1}{8}u_{j-1}^n + \frac{3}{4}u_j^n + \frac{3}{8}u_{j+1}^n$$

- (a) State the condition on the coefficients  $c_k$  for a linear scheme to be **monotonicity-preserving**. Which of the two schemes satisfies this condition?
- (b) For a monotone step  $u_{-1}^0 = 1, u_0^0 = 1, u_1^0 = 0, u_2^0 = 0$ , compute  $u_0^1$  and  $u_1^1$  using Scheme B. Does Scheme B create an unphysical overshoot ( $> 1$ ) or undershoot ( $< 0$ )?
- (c) State **Godunov's Barrier Theorem (1959)** in one or two clear sentences.